Name:Pirasana Vinayagam Manohar Register No: 250701516
Experiment Name:Feature Scaling

```
import numpy as np
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler,
MinMaxScaler

df = pd.read_csv('pre-process_datasample.csv')

df['Country'] = df['Country'].fillna(df['Country'].mode()[0])

features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values

age = SimpleImputer(strategy="mean", missing_values=np.nan)
salary = SimpleImputer(strategy="mean", missing_values=np.nan)

features[:, 1:2] = age.fit_transform(features[:, 1:2])
features[:, 2:3] = salary.fit_transform(features[:, 2:3])

oh = OneHotEncoder(sparse_output=False)
Country = oh.fit_transform(features[:, 0:1])

final_set = np.concatenate((Country, features[:, 1:3]), axis=1)

sc = StandardScaler()
feat_standard_scaler = sc.fit_transform(final_set)
print(feat_standard_scaler)

mms = MinMaxScaler(feature_range=(0, 1))
feat_minmax_scaler = mms.fit_transform(final_set)
print(feat_minmax_scaler)
```

```
[[ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01  7.58874362e-01
   7.49473254e-01]
 [-8.16496581e-01 -6.54653671e-01  1.52752523e+00 -1.71150388e+00
  -1.43817841e+00]
 [-8.16496581e-01  1.52752523e+00 -6.54653671e-01 -1.27555478e+00
  -8.91265492e-01]
 [-8.16496581e-01 -6.54653671e-01  1.52752523e+00 -1.13023841e-01
  -2.53200424e-01]
 [-8.16496581e-01  1.52752523e+00 -6.54653671e-01  1.77608893e-01
   6.63219199e-16]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01 -5.48972942e-01
  -5.26656882e-01]
 [-8.16496581e-01 -6.54653671e-01  1.52752523e+00  0.00000000e+00
  -1.07356980e+00]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01  1.34013983e+00
   1.38753832e+00]
```

```
 [-8.16496581e-01  1.52752523e+00 -6.54653671e-01  1.63077256e+00
   1.75214693e+00]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01 -2.58340208e-01
   2.93712492e-01]]
[[1.         0.         0.         0.73913043 0.68571429]
 [0.         0.         1.         0.         0.        ]
 [0.         1.         0.         0.13043478 0.17142857]
 [0.         0.         1.         0.47826087 0.37142857]
 [0.         1.         0.         0.56521739 0.45079365]
 [1.         0.         0.         0.34782609 0.28571429]
 [0.         0.         1.         0.51207729 0.11428571]
 [1.         0.         0.         0.91304348 0.88571429]
 [0.         1.         0.         1.         1.        ]
 [1.         0.         0.         0.43478261 0.54285714]]
```